

A Data Analytic Report on India's Performance Grading Index (PGI 2.0) 2023–2024

1. Project Overview and Objective

This project focuses on analysing India's Performance Grading Index (PGI 2.0) dataset of the academic year 2023–24. The Performance Grading Index (PGI) framework evaluates states and union territories based on their school education performance across multiple dimensions.

Objective:

- To clean, process, and analyse the PGI dataset using Excel and Power BI.
- To visualize state-wise, category-wise, and regional educational performance.
- To identify the strongest and weakest performance areas under the PGI 2.0 structure.
- To present actionable insights that can help policymakers improve education outcomes.

2. Data Source

Source: Ministry of Education, Government of India – PGI 2.0 dataset

Link: <https://www.data.gov.in/resource/stateut-wise-district-score-during-2023-24>

Format: Excel (.xlsx)

Coverage: 28 States and 8 Union Territories

Time Period: Academic Year 2023–24

Domain: Education

3. Problem Statement

To evaluate India's educational performance using PGI 2.0 indicators by:

- Measuring category-wise and overall performance of States/UTs.
 - Identifying top-performing and underperforming states.
 - Comparing district grades and ratings
 - Analyzing regional variations and improvement areas.
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4. Attribute (Column) Details

Attribute Name	Data Type	Description
State/UT	Text	Name of the State or Union Territory
District	Text	District name within each State/UT
Outcomes	Numeric	Measures learning outcomes and academic results
ECT (Effective Classroom Transactions)	Numeric	Reflects quality and efficiency of classroom teaching
IF&SE (Infrastructure, Facilities & School Environment)	Numeric	Represents physical infrastructure and learning environment
SS&CP (School Safety & Child Protection)	Numeric	Indicates safety, inclusivity, and welfare policies for students
DL (Digital Learning)	Numeric	Evaluates ICT and technology integration in schools
GP (Governance Processes)	Numeric	Measures effectiveness of school management and planning
Overall	Numeric	Total PGI score (sum of all categories; maximum = 600)
Overall %	Numeric	Percentage calculated from overall score
Rating	Text	Qualitative assessment (Excellent, Good, Fair, Poor)
Grade	Text	Performance grade category (Uttam, Pracheshta, Akanshi)
Region	Text	Geographic region classification (North, South, East, West, Central, North-East)
State/UT Type	Text	Distinguishes between State and Union Territory

5. Grade Classification and Explanation

Grade Name	Description / Meaning	Score Range
Uttam-2	Represents the highest level of performance excellence across all indicators.	Above 400
Pracheshta-1	Districts showing commendable performance but still below top-tier levels.	300 and above
Pracheshta-2	Moderate achievers with potential for improvement.	250–299
Pracheshta-3	Developing category with partial compliance; needs enhanced support.	200–249
Akanshi-1	Represents the lowest performance level where immediate attention is needed.	Below 200

6. Tools & Technologies

Tool	Purpose
Excel	Data cleaning, descriptive analysis, pivot tables, ranking, and overall % calculations
Power Query (Power BI)	Data transformation, creation of calculated columns, and regional classification
Power BI Desktop	Interactive dashboard creation, data modelling, DAX calculations, bookmarks, and insights visualization

7. Data Preprocessing (Excel + Power Query)

- ~ Cleaned dataset (removed duplicates, standardized formats)
 - ~ Conducted descriptive analysis for overall performance and Categories of PGI
 - ~ Added calculated fields for Overall%, Rating, State-Region, Rank in Power Query.
 - ~ Conditional Formatting for Overall % using icon sets
 - ~ Used Conditional Columns in Power Query for Region mapping.
 - ~ Derived insights using Pivot Tables for the following
 - category-wise top 10 performing states,
 - District count,
 - State Region vs Overall% and
 - Rating
 - Overall Grade Rating Vs Districts
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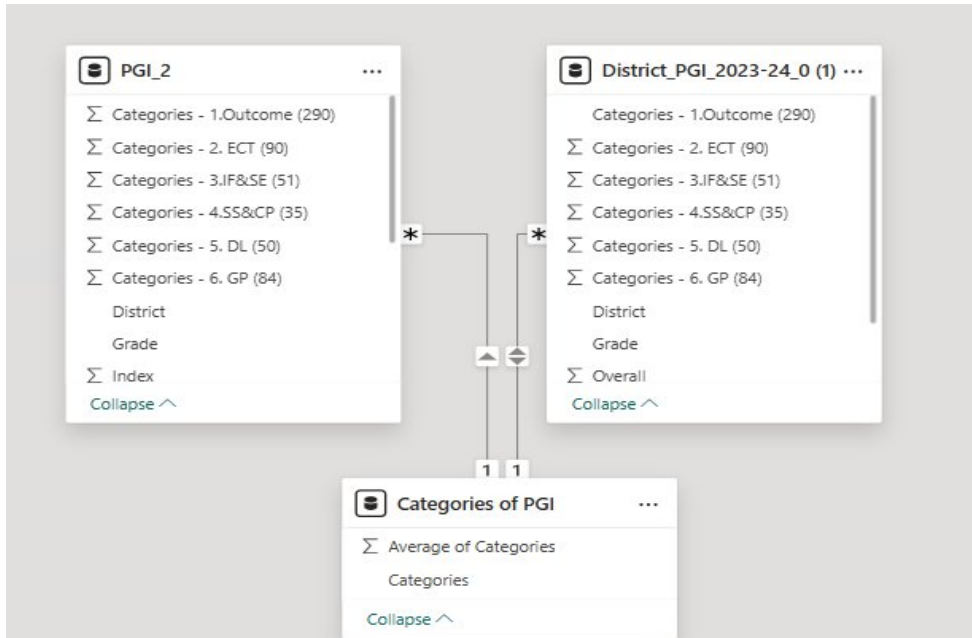
8. Data Modelling and DAX in Power BI

Manually created a table for categories with their averages for ease of visualization

Categories	Average of Categories
Outcomes	123.13
Effective Classroom Transactions(ECT)	56.04
Infrastructure, Facilities & Student Entitlements (IF&SE)	28.31
School Safety and Child Protection (SS&CP)	16.02
Digital Learning(DL)	16.05
Governance Processes (GP)	48.63

DAX Measures: Average Overall

Relationship in model view



9. Dashboard Pages and Visualizations

Commonly used visuals in all pages: Page Navigator, Slicers and Clear slicer button

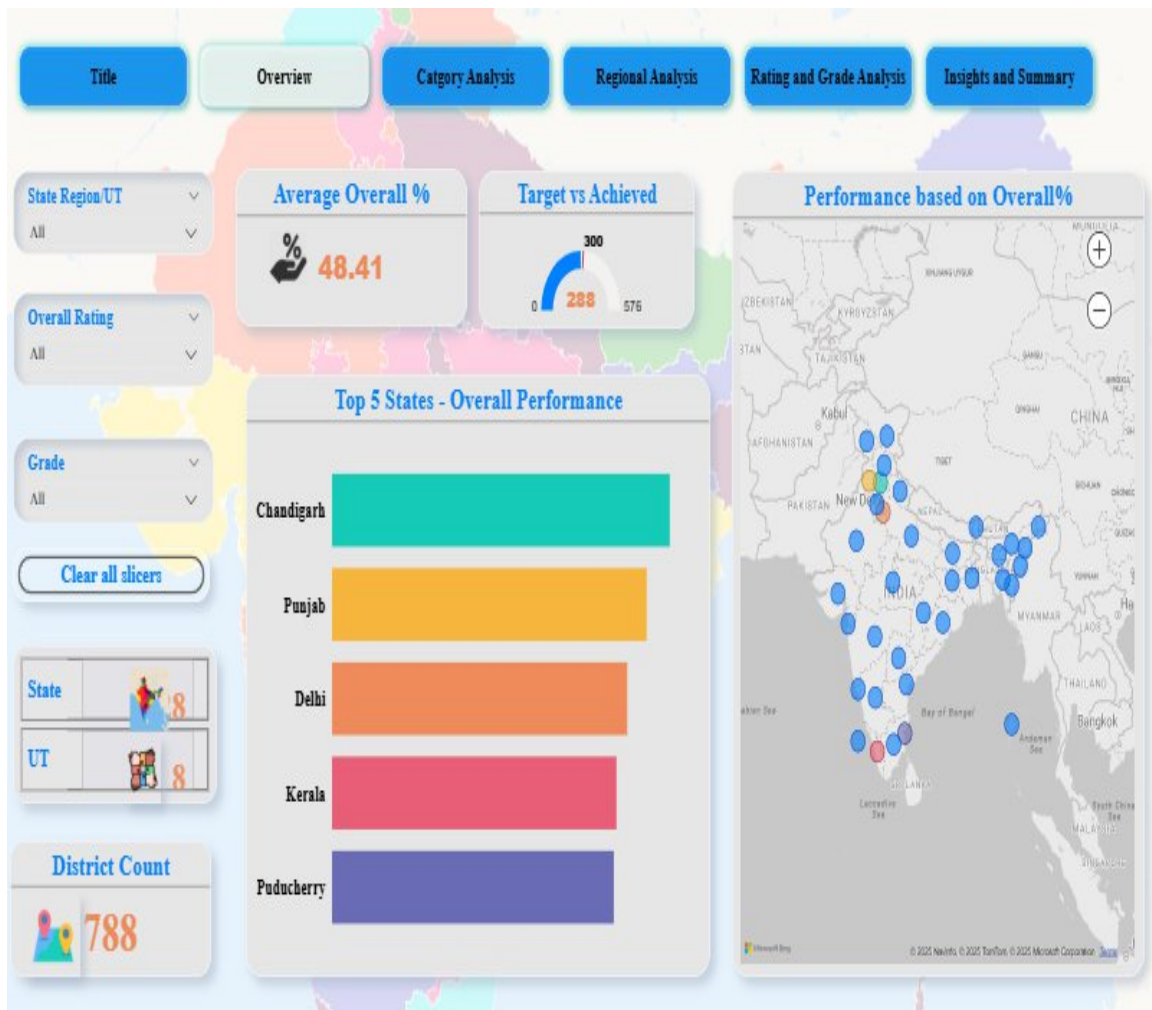
Page	Visuals Used	Purpose
Overview Dashboard	Card (New), Card (Normal), Clustered Bar Chart, Gauge, Map,	Gives national-level performance snapshot
Category Analysis	Clustered Column Chart, Pie-Chart, Decomposition Tree, TreeMap, Q&A and Cards	Highlights strong and weak performing PGI categories
Grade & Rating Analysis	Matrix Table, Table, Pie-Chart, Clustered Column Chart, Clustered Bar Chart and Card (Normal)	Shows district spread across grades
Regional Analysis	Stacked Column Chart, Map, Pie-Chart, TreeMap and Card (Normal).	Compares regional differences
Insights & Recommendations	Text visuals	Summarizes findings and improvement areas

REPORT SNAPS

1.Title Page Snap



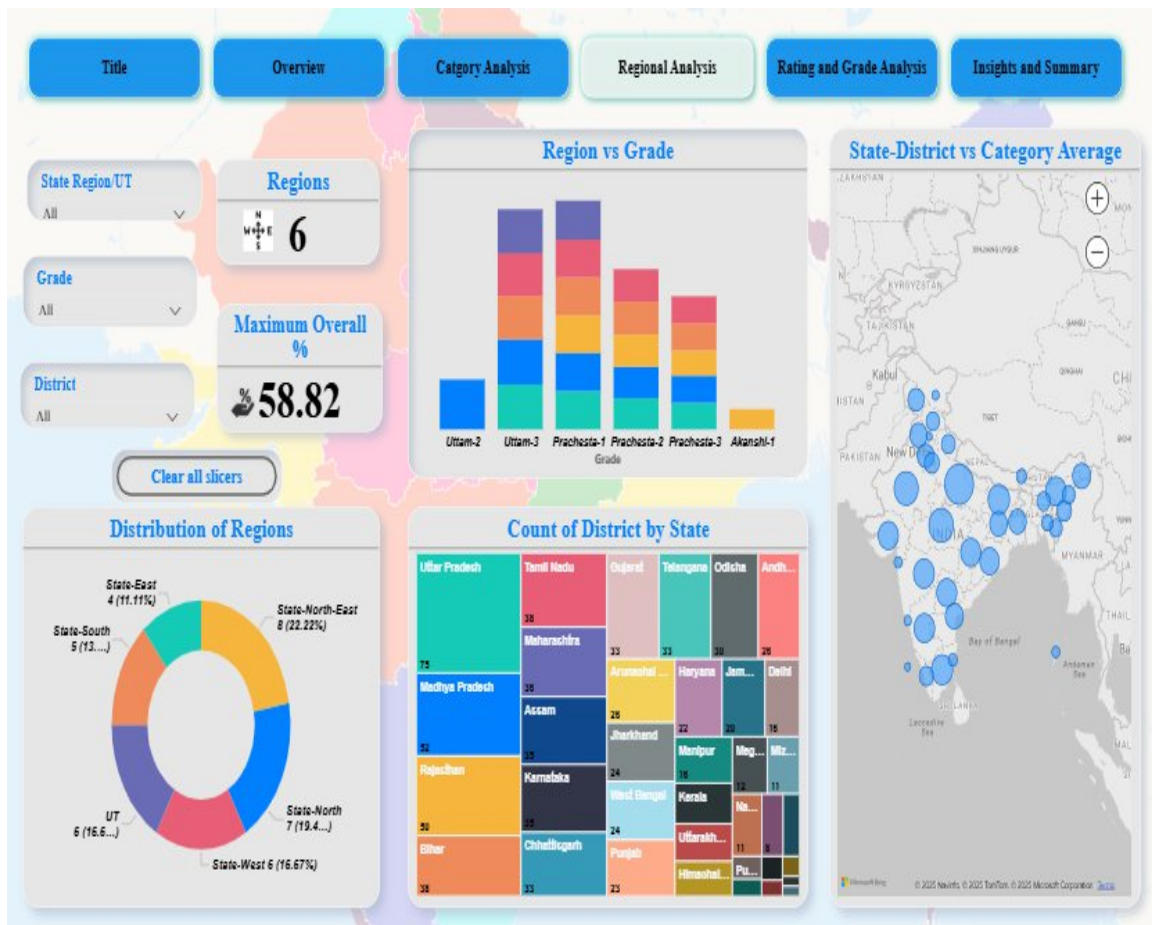
2. Overview Snap



3.Category Analysis Snap



4.Regional Analysis Snap



5. Grade & Rating Analysis Snap



5.Insights & Summary



10. Insights and Recommendations

- Outcomes and Governance Processes categories are leading contributors to PGI performance.
- Digital Learning and School Safety remain underperforming areas needing investment.
- Majority of districts fall under Pracheshta grades, showing steady but incomplete progress.
- Northern and Southern regions outperform Central and Eastern India.
- Encouraging mentorship between Uttam and Pracheshta states can help balance growth.

Descriptive Insight Most states fall under Pracheshta levels, indicating moderate achievement in overall PGI performance.

Diagnostic Insight: Digital Learning and School Safety are the weakest categories, which is pulling down the overall score across multiple regions.

Predictive Insight: If Governance and Outcomes continue improving at the same pace, more states are likely to move from Pracheshta into Uttam tier in the next evaluation cycle.

Prescriptive Insight: Focus on strengthening digital infrastructure and safety initiatives with targeted interventions to uplift low-performing districts quickly

11. Conclusion

This mini project showcases how data analytics and visualization can make large-scale educational data actionable. By using Excel for preprocessing and Power BI for interactive visualization, meaningful insights were extracted from the PGI dataset. The study emphasizes that while progress is evident across most states, improving digital learning and safety components can help India achieve true educational excellence.
